

BUILDING THE HUMAN CIRCULATORY SYSTEM MODEL

Interactive Hands-On Science Lesson Guide & Activity Plan

1. Warm-Up Activity: Feel Your Heartbeat!

Activity: Finding Your Pulse

Place two fingers gently on your wrist or the side of your neck. Can you feel a steady, rhythmic thumping?

- **How It Works:** Your heart acts like a continuous, non-stop water motor pumping blood throughout your entire body.
- **Pulse Recheck Challenge:** Jump on the spot for 30 seconds! Check your pulse again. Notice how much faster and harder your heart is pumping to deliver energy and oxygen to your working muscles.

2. Meet "Raju Red-Cell" & Core Concepts

Follow the journey of *Raju Red-Cell*, a red blood cell traveling through your body in less than a single minute!

Component / Concept	Biological Role	Real-World Analogy
Red Blood	Oxygenated blood (full of fresh oxygen from the lungs).	Refueled train carrying essential goods.
Blue Blood	Deoxygenated blood (oxygen used up by the body).	Empty train returning to get refueled.
Blood Vessels	Highways and tubes connecting every organ and cell.	Indian Railways connecting every city across the nation.
Lungs	Exchanges carbon dioxide for fresh oxygen.	A CNG / Fuel Station refueling the red blood cells.
Heart Valves	Flaps that prevent blood from flowing backward.	Metro Turnstiles that only let passengers pass ONE way.

The Continuous Loop Journey:
Body → Heart → Lungs (Oxygen Refuel) → Heart → Body (Round & Round!)

3. Hands-On Activity: Let's Get Building!

Build your working hands-on model of the human circulatory system using everyday laboratory and household materials.

Materials Needed:

- **2 Balloons:** One stretched as a flexible pumping diaphragm/membrane over the chamber, and one cut to act as a one-way heart valve.
- **1 Plastic Box (with tight lid):** Represents the heart chamber / chest cavity holding blood.
- **Clear Flexible Tubing:** Represents the blood vessels carrying fluid into and out of the heart.
- **Coloured Water (Red & Blue):** Represents oxygenated and deoxygenated blood traveling through the system.
- **Cardboard body silhouette template, tape/zip-ties, and scissors.**

Step-by-Step Construction Guide:

Step 1: Chamber & Tubing Assembly

Fill the plastic box halfway with coloured water. Carefully pierce two small holes through the lid of the plastic box. Insert two lengths of clear flexible tubing into the holes so one reaches deep into the coloured water (intake/artery) and the second acts as the outlet tube.

Step 2: Attach Balloon Diaphragm & Valve

Cut the neck off *Balloon #1* and stretch the rubber membrane tightly over the top of the plastic box lid to create an airtight seal (acting as the contracting heart muscle/diaphragm). Take *Balloon #2*, snip off its narrow tip, and tape it over the outer end of the outlet tube to form a flexible, one-way valve flap preventing backflow.

Step 3: Mount, Pump & Observe

Secure your plastic box chamber and flexible tubing onto the cardboard silhouette using tape or zip ties. Press down rhythmically on the balloon membrane covering the plastic box! Watch how the increased pressure pushes coloured water up through the clear tubing blood vessels, demonstrating how heart muscle contractions pump blood across the body.

Top Tip: Work in pairs! Ask a partner to hold the plastic box steady while you stretch the balloon diaphragm and attach the tubing and valve.

STUDENT PRACTICE WORKSHEET

Human Circulatory System Model — Test Your Knowledge

MODEL CONSTRUCTION CHECKLIST & MATERIALS USED:

- **2 Balloons** (Pump Diaphragm & One-Way Valve) • **1 Plastic Box** (Heart Chamber)
- **Clear Flexible Tubing** (Blood Vessels) • **Coloured Water** (Red & Blue Blood)

Instructions: Answer the following questions based on the lesson and model activity.

Question 1 (Multiple Choice)

What color is used to represent oxygenated blood in medical diagrams and in our model?

- ☐ A) Blue
- ☐ B) Red
- ☐ C) Yellow
- ☐ D) Green

Question 2 (Multiple Choice)

In our hands-on model, which component acted as the flexible muscle that creates pressure to pump the coloured water through the tubing?

- ☐ A) Clear tubing
- ☐ B) Cardboard silhouette
- ☐ C) Stretched balloon membrane on the plastic box
- ☐ D) Red food coloring

Question 3 (Short Answer)

Why does your heart beat faster when you jump or run for 30 seconds compared to when you are resting?

Question 4 (Analogy Matching)

Match each part of the human circulatory system to its corresponding model part and real-world analogy:

Circulatory System Part	Model Element (Balloons, Plastic Box, Tubing, Coloured Water)	Real-World Analogy
1. Heart Chamber & Muscle	Plastic Box & Stretched Balloon Membrane	Water Motor / Pump
2. Blood Vessels	Clear Flexible Tubing	Railway Network / Highways
3. Heart Valve	Cut Balloon Flap on Tube	Metro Turnstile
4. Oxygenated / Deoxygenated Blood	Coloured Water (Red / Blue)	Refueled / Empty Freight Trains

Question 5 (Flow Order Sequence)

Arrange the following stages of Raju Red-Cell's journey in the correct chronological loop (Number 1 to 4):

- [] Blood travels to the body to deliver oxygen
- [] Deoxygenated blood returns to the heart
- [] Blood goes to the lungs to pick up oxygen
- [] Oxygenated blood enters the heart to be pumped out

Question 6 (Model Reflection)

When building your model with the plastic box, tubing, and balloon flap valve, what would happen if the balloon flap valve was missing or placed backward? Explain how this affects the direction of coloured water flow.